

a striking result: when the relevant articles have low reliability, social media platforms that aim to maximize engagement will tend to design algorithms that increase homophily and create filter bubbles, propagating misinformation. Intuitively, high-reliability content tends to spread anyway because most users recognize it as such and share it, and low homophily contributes to its spread by increasing its circulation throughout the network. In contrast, low-reliability content will be ignored or disliked by agents who disagree with its message and believe it to contain misinformation. Engagement with low-reliability content can be increased if the platform ensures that it remains among users ideologically aligned with its message, who would be willing to share it with like-minded others without fear of being called out by users with different ideologies. Hence, creating filter bubbles becomes an attractive strategy for engagement-maximizing platforms. It is particularly troublesome that such filter bubbles are created precisely when the relevant content is low-reliability and likely to contain misinformation.

If platform algorithms are propagating misinformation, can public policy counter and discourage this type of behavior?<sup>4</sup> The answer is yes, but with some caveats. In the last part of the paper, we discuss four different types of regulatory policies, and in each case, we show how they may reduce misinformation but also point out the possibility that, if they are not designed well, they can backfire and exacerbate the problem.

First, we discuss regulations that force platforms to reveal the provenance of articles, making it easier for users to identify falsehoods (e.g., claims originating from less reputable sources, such as InfoWars). Though generally useful, such provenance information can backfire when it instills a sense of “false security” that the content has been verified by previous social media users. Second, we look at potential censorship of articles identified by a regulator as likely containing misinformation. While censorship can help reduce the viral spread of misinformation, it also generates an “implied truth” effect (Pennycook et al. (2020a)) that contributes to the viral spread of questionable content that escapes censorship. Third, we discuss “performance targets”, where the regulator places limits on the amount of misinformation that circulates on the platform. Such targets tend to better align user and regulator preferences, but unless they appropriately target sensational content, platforms can meet the required performance limits by focusing on less sensational material that was unlikely to become viral in any case, and the least reliable content may still continue to circulate virally. Lastly, we show how direct regulation of platform algorithms can reduce misinformation, but also point out that the non-monotone effects of homophily imply that such regulations need to be finely calibrated.

**Related Literature.** Our paper builds on a large body of work on models of misinformation. In addition to the literature mentioned previously, several other papers in this literature are related to our findings.

Much previous work has focused on the susceptibility of boundedly-rational agents to engage with misinformation. In Acemoglu et al. (2010) and Acemoglu et al. (2013), the existence of persuasive agents can impede information aggregation and enable misinformed beliefs to survive, and sometimes even become dominant, in the population. In Mostagir et al. (2021) and Mostagir and Siderius (2021),

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<sup>4</sup>As of August 2021, federal law protects social media platforms from being held responsible for content posted by its users (see Section 230 of the Communications Decency Act of 1996, discussed by <https://hbr.org/2021/08/its-time-to-update-section-230>).